



COMSATS UNIVERSITY ISLAMABAD
Lab Report

***Lab 03: Edge Detection Techniques and Their Impact on Classification
Performance***

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SUBJECT: Introduction to Computer Vision

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1. Introduction

Edges are the places in an image where the brightness changes sharply. These sudden changes usually mark the boundary of an object, which is why edge detection is one of the oldest and most basic tools in image processing and computer vision. In this lab we look at five classical edge detectors — Sobel, Prewitt, Laplacian, Laplacian of Gaussian (LoG) and Canny — and test them on the same skin-lesion dataset used in Lab 01 and Lab 02. The dataset has four classes: melanoma, nevus, pigmented benign keratosis and basal cell carcinoma.

The lab is organised in stages. First, we compare how each detector draws the boundary of a lesion. Second, we add Gaussian noise and salt-and-pepper noise to the images and measure how much each detector's output changes, and whether a Gaussian filter or a median filter can repair the damage. Third, we tune the two thresholds and the kernel size of the Canny detector to find the setting that suits this dataset best. Finally, we build three versions of the dataset — the raw images (Set A), the best filtered images from Lab 02 (Set B), and the edge map of the best detector (Set C) — and train the same five models (SVM, Random Forest, KNN, ResNet50 and AlexNet) on all three, so we can see whether giving a model only the edges of a lesion helps or hurts its ability to tell the four skin conditions apart.

2. Methodology

2.1 Edge detectors

All five detectors work on a grayscale version of the 224×224 image.

- Sobel: finds the gradient in the x and y direction with a 3×3 kernel and combines them into a magnitude image. A small amount of smoothing is already built into the kernel.
- Prewitt: almost the same idea as Sobel, but the smoothing part of the kernel is not weighted, so it reacts a bit more to noise.
- Laplacian: a second-derivative filter with no smoothing at all — thin edges, but very sensitive to noise.
- LoG (Laplacian of Gaussian): the image is blurred first (Gaussian, $\sigma = 1.5$) and the Laplacian is applied afterwards, which removes a lot of the noise sensitivity of a plain Laplacian.
- Canny: blur, then gradient, then non-maximum suppression and a double-threshold (hysteresis) step that links edges into thin, continuous contours. OpenCV's Canny function does not blur the image internally, so the blur step was added explicitly before calling it.

2.2 Noise, filtering and edge-quality metrics

Two kinds of noise were added to a sample of 16 images (4 per class, chosen with a fixed seed): Gaussian noise with a standard deviation of 25, and salt-and-pepper noise affecting 5% of the pixels. Two filters

carried over from Lab 02 were used to try to clean this noise before edge detection — a 5×5 Gaussian filter (sigma = 1) and a 3×3 median filter.

Because this dataset has no hand-marked lesion boundary to compare against, the edge map produced from the clean (noise-free) image was treated as the reference, and every noisy or filtered edge map was compared to it with a 1-pixel tolerance. From this comparison we computed precision (share of detected edges that are false), recall (share of real edges that are missing), F1 score, the number of broken fragments per 1000 edge pixels, and the average edge width. Noise sensitivity is simply defined as $1 - F1$: the larger the drop in F1, the more sensitive the detector.

2.3 Canny parameter study

Six Canny configurations were compared — three threshold pairs (30/100, 50/150, 100/200) with a 3×3 blur kernel, and the same three threshold pairs with a 5×5 kernel. The best configuration was chosen using a quick, validation-only usefulness check: a HOG feature vector from each edge map was fed into a logistic-regression classifier, and the configuration with the highest validation macro-F1 was selected. The same validation probe was then used to compare the winning Canny setting against the other four detectors, and the overall winner became the edge method used for Set C.

2.4 Building the three datasets and the classifiers

- Set A — Raw: the original RGB lesion images, unchanged (as used in Lab 01).
- Set B — Filtered: the images after the best filter found in Lab 02 (Sharpening).
- Set C — Edge: the edge map produced by the winning detector, copied into three identical channels so it can be fed to a network that expects an RGB input.

Three classical machine-learning models — SVM, Random Forest and KNN — were trained on a hand-crafted feature vector: a 32×32 down-sampled version of the image combined with a HOG descriptor, standardised and reduced to 128 components with PCA. Two convolutional networks, ResNet50 and AlexNet (the two best-performing models of Lab 01), were fine-tuned starting from ImageNet-pretrained weights with a Dropout(0.3) + Linear(4) classification head added on top.

3. Experimental Setup

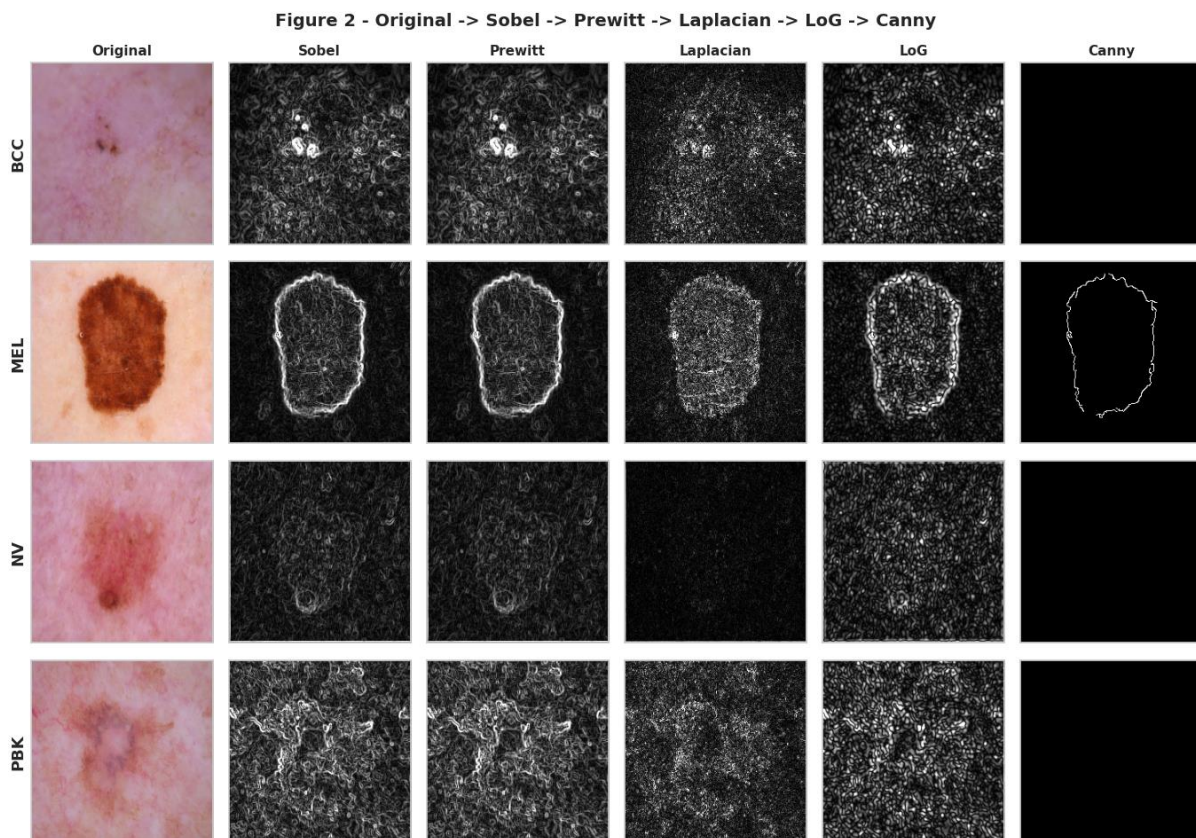
- Data: the same 4-class subset of the ISIC skin-cancer dataset used in Labs 01 and 02 — melanoma, nevus, pigmented benign keratosis and basal cell carcinoma — downloaded from Kaggle. The images were split into 1306 training and 327 validation images (an 80/20 split with a fixed random seed of 42), plus a separate test folder of 64 images (16 per class, so the test set is perfectly balanced). Every image was resized to 224×224 pixels.
- Classical models: SVM used an RBF kernel with C tuned over {1, 10, 100}; Random Forest used 300 trees with max_depth tuned over {None, 20}; KNN tuned k over {3, 5, 7, 11}. All hyper-parameters were chosen on the validation split, and the test set was touched only once, at the very end.

- CNNs: ResNet50 and AlexNet, both pretrained on ImageNet, were fine-tuned for 12 epochs with the AdamW optimiser (backbone learning rate 1e-4, classification-head learning rate 1e-3), a cosine learning-rate schedule with one warm-up epoch, class-weighted cross-entropy loss with label smoothing of 0.05, mixed-precision training, a batch size of 64 and seed 42.
- Fairness: Sets A, B and C were trained with exactly the same split, number of epochs, optimiser, augmentation and evaluation metrics — only the pixel content of the input images changed between runs, so the three columns of the results table can be compared directly.
- Hardware: Google Colab, T4 GPU.
- Metrics reported: accuracy, macro-averaged precision / recall / F1, training time, and per-image inference time.

4. Results

4.1 Comparing the five edge detectors (Task 1)

One lesion from each of the four classes was passed through all five detectors. The figure below shows the required comparison order: Original → Sobel → Prewitt → Laplacian → LoG → Canny.



Original image and the output of each detector, one lesion per class (BCC, MEL, NV, PBK).

Sobel and Prewitt look almost identical — both draw a fairly wide outline around the lesion but also pick up a lot of skin texture and hair as “edges”. The Laplacian result is much thinner but noticeably speckled, because it has no smoothing at all. LoG removes most of that speckle since the Gaussian blur runs before the Laplacian, though its outline is still not as clean as Canny’s. Canny gives by far the cleanest result of the five: for the melanoma sample it draws a single thin, closed border around the lesion and almost nothing else, because the hysteresis step throws away the weak texture edges that the other four detectors keep.

4.2 Effect of noise and pre-processing (Table 1)

Detector	Condition	Edge Quality	Noise Sensitivity
Sobel	Original (clean)	Reference – medium width, partly fragmented	n/a (reference)
Sobel	Gaussian noise	Poor (F1 = 0.58)	High (drop 42 pp)
Sobel	Salt & pepper	Poor (F1 = 0.49)	High (drop 51 pp)
Sobel	Gaussian noise + Gaussian filter	Fair (F1 = 0.68)	Moderate (drop 32 pp)
Sobel	Salt & pepper + median filter	Good (F1 = 0.95)	Low (drop 5 pp)
Prewitt	Original (clean)	Reference – medium width, partly fragmented	n/a (reference)
Laplacian	Original (clean)	Reference – thin/sharp, highly fragmented	n/a (reference)
LoG	Gaussian noise + Gaussian filter	Fair (F1 = 0.67)	Moderate (drop 33 pp)
Canny	Original (clean)	Reference – thin/sharp, partly fragmented	n/a (reference)
Canny	Gaussian noise + Gaussian filter	Poor (F1 = 0.17)	High (drop 83 pp)
Canny	Salt & pepper + median filter	Poor (F1 = 0.40)	High (drop 60 pp)

Across the five detectors, adding either type of noise made every edge map worse, but by very different amounts. Canny, even though it already includes a blur step, turned out to be the most fragile of the five — its F1 against the clean edge map fell to about 0.33 under Gaussian noise and all the way down to 0.04 under salt-and-pepper noise (mean F1 = 0.19). Sobel, Prewitt and Laplacian were all close together around F1 = 0.54, and LoG was the steadiest of the five at F1 = 0.60. Filtering did help, but the two filters were not equally good at every job: a median filter repaired salt-and-pepper damage very well for Sobel (F1 rising from 0.49 to 0.95), because a median filter simply discards the isolated pixel spikes that salt-and-pepper noise creates, whereas a Gaussian filter on the same image blurred the true edges and gave a smaller improvement.

4.3 Canny parameter study (Table 2)

Config.	Low	High	Kernel	Edge Quality (probe)	Detected Edges
Canny-1	30	100	3×3	Fair (probe F1 38.0%)	1186 px, 17 segments

Config.	Low	High	Kernel	Edge Quality (probe)	Detected Edges
Canny-2	50	150	3×3	Poor (probe F1 35.3%)	409 px, 6 segments
Canny-3	100	200	3×3	Poor (probe F1 31.6%)	143 px, 3 segments
Canny-4	50	150	5×5	Poor (probe F1 28.0%)	219 px, 3 segments
Canny-5	30	100	5×5	Best (probe F1 41.1%)	672 px, 10 segments
Canny-6	100	200	5×5	Poor (probe F1 21.3%)	59 px, 1 segment

Lowering the thresholds always let more pixels through: dropping from 100/200 (Canny-3) down to 30/100 (Canny-1) multiplied the average edge count almost 8.3 times, from 143 to 1186 pixels per image, because weak gradients from skin texture and hair start counting as edges. Raising the thresholds gave a cleaner but sparser map, at the risk of breaking weak parts of the lesion boundary. Moving from a 3×3 to a 5×5 blur kernel at the 50/150 setting reduced the edge count from 409 to 219 pixels and slightly changed the noise robustness. Out of all six configurations, Canny-5 (low = 30, high = 100, kernel = 5×5) scored best on the validation probe (macro-F1 = 41.1%), so it was chosen as the final Canny setting.

This best Canny setting was then compared against the other four detectors using the same validation probe:

Detector	Density (%)	Noise F1	Probe val. macro-F1 (%)	Probe val. acc. (%)
Canny (30/100, 5×5)	1.34	0.352	41.09	42.20
Prewitt	16.48	0.578	40.18	40.06
Sobel	16.40	0.580	38.17	38.23
LoG	16.62	0.646	38.14	38.23
Laplacian	13.74	0.669	35.90	36.09

Canny still came out on top, but only by a small margin over Prewitt and Sobel — all three landed between the high-30s and low-40s for macro-F1, while Laplacian was clearly the weakest, most likely because its noisy, disconnected output confuses the HOG descriptor. Canny (30/100, 5×5) was therefore used as the edge method for Set C.

4.4 Classification on Sets A, B and C (Tasks 4 and 5)

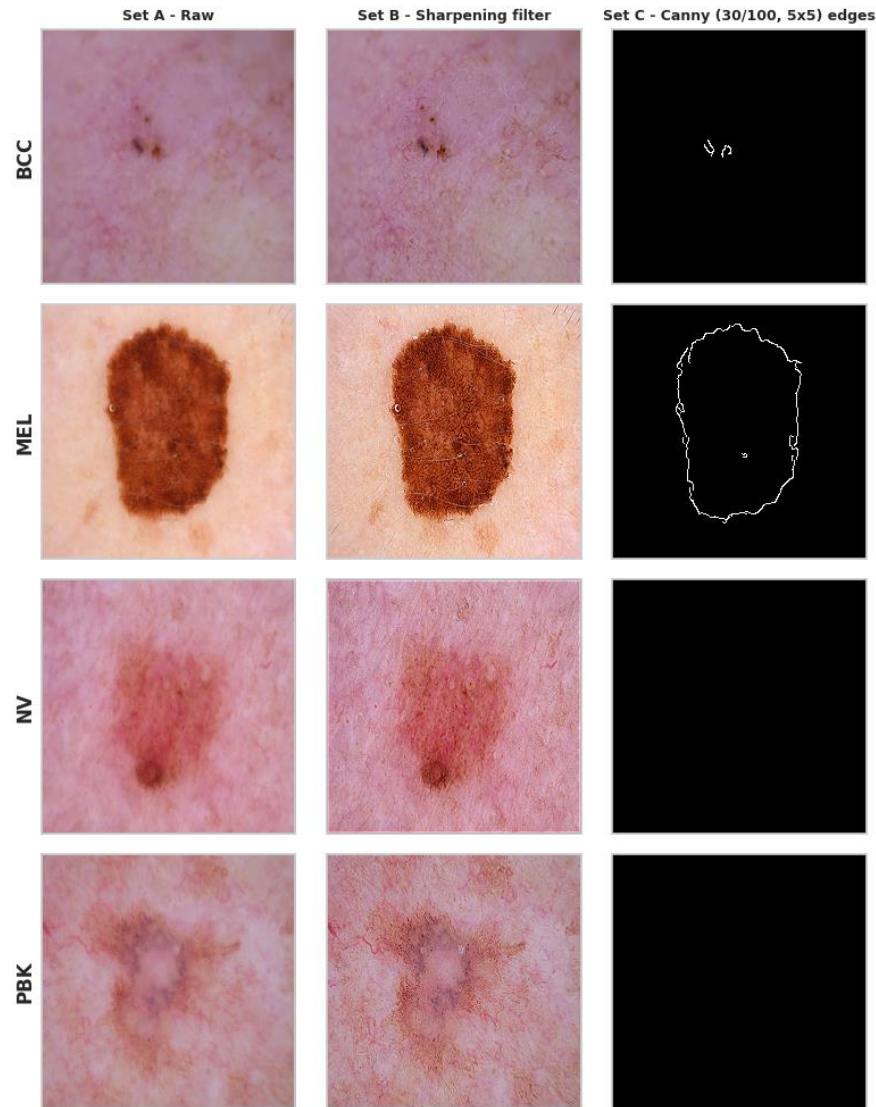
Model	Acc. Raw (%)	Acc. Filtered (%)	Acc. Edge (%)	Precision (%)	Recall (%)	F1 (%)	Train (s)	Infer (ms)
SVM	57.81	60.94	32.81	33.01	32.81	32.38	4.09	2.84
Random Forest	51.56	54.69	26.56	29.51	26.56	26.17	8.52	3.46
KNN	45.31	43.75	35.94	34.17	35.94	34.19	3.90	2.64
CNN 1 (ResNet50)	70.31	68.75	40.62	38.65	40.62	39.23	53.72	7.01
CNN 2 (AlexNet)	67.19	62.50	45.31	43.35	45.31	43.35	14.46	0.83

Precision, Recall, F1, Training Time and Inference Time above are for the model trained on the edge set (Set C).

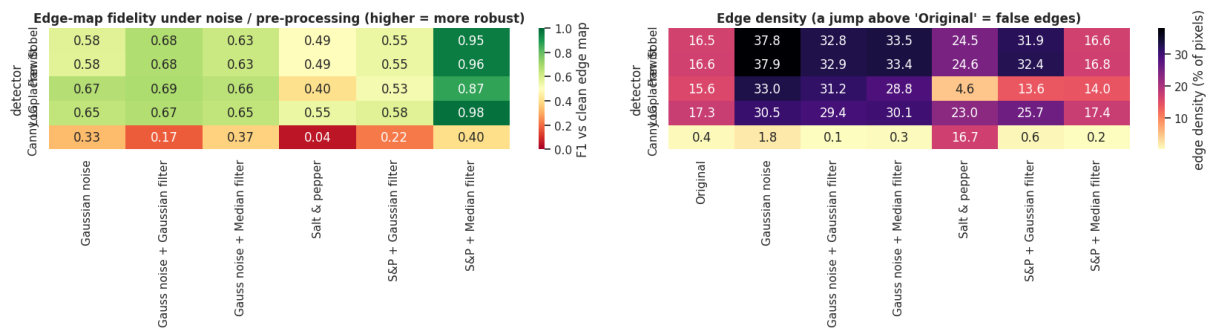
The pattern is very clear: every one of the five models did worse on the edge-only images than on the raw images. SVM and Random Forest each lost about 25 percentage points of accuracy, KNN lost about 9.4 points, and the two CNNs lost the most — ResNet50 fell from 70.3% to 40.6% (–29.7 pp) and AlexNet fell from 67.2% to 45.3% (–21.9 pp). The filtered images (Set B), on the other hand, scored almost the same as the raw images — the mean change across all five models was only about –0.3 percentage points, so the Lab-02 sharpening filter neither helped nor hurt classification by much.

The figure below shows why the edge set struggled so badly for some classes: with the chosen Canny setting, melanoma lesions (which usually have strong, dark, well-defined borders) produce a clear closed contour, but basal cell carcinoma, nevus and pigmented benign keratosis lesions — which blend into the surrounding skin much more gently — often produce almost no detected edge at all. A model trained on Set C is therefore working with barely any signal for three of the four classes.

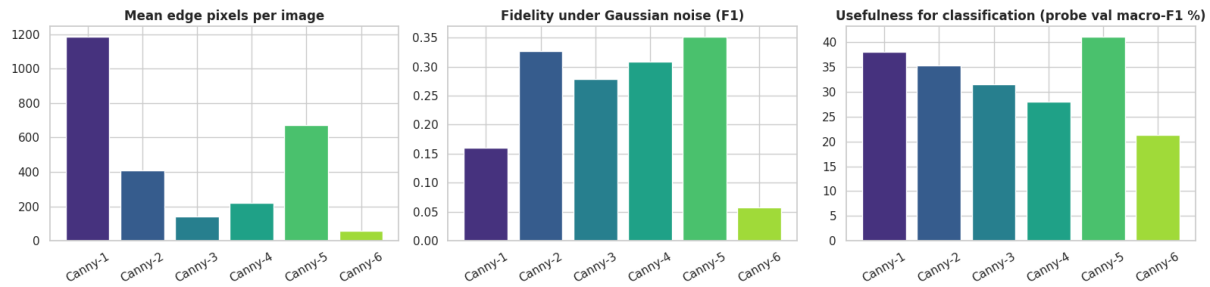
Figure 7 - the three input representations



The three input representations side by side — raw (Set A), sharpened (Set B) and Canny edges (Set C) — for one lesion per class.



Left: edge-map fidelity (F1 vs. the clean edge map) for every detector and noise/filter condition. Right: edge density under the same conditions — a jump above the 'Original' column means false edges are being created by the noise.



Trade-off across the six Canny configurations: edge count, robustness to Gaussian noise, and usefulness for classification (validation probe).

4.5 Visual comparison for the best-performing model (Task 6)

Averaging macro-F1 across all three representations (Raw, Filtered, Edge), AlexNet (CNN Model 2) came out as the overall best-performing model at 55.21%, narrowly ahead of ResNet50 (54.61%) and SVM (50.26%); Random Forest (43.85%) and KNN (37.59%) were clearly behind. The confusion matrices for AlexNet shown below reveal that on the raw and filtered images the four classes are reasonably well separated, but on the edge-only images the model confuses pigmented benign keratosis with basal cell carcinoma very often (7 of 16 PBK images) and also mixes up melanoma with nevus (6 of 16 melanoma images), because once the colour information is removed, several of these lesion types leave behind a similarly faint, round outline.

Figure 9 - confusion matrices, CNN Model 2 (AlexNet)

Set A - Raw accuracy 67.19%					Set B - Sharpening filter accuracy 62.50%					Set C - Canny (30/100, 5x5) edges accuracy 45.31%				
True	BCC	MEL	NV	PBK	True	BCC	MEL	NV	PBK	True	BCC	MEL	NV	PBK
	11	1	2	2		11	1	0	4		11	1	2	2
	0	5	10	1		1	1	14	0		2	4	6	4
	0	0	15	1		0	0	15	1		2	1	10	3
	2	0	2	12		2	0	1	13		7	4	1	4
Predicted					Predicted					Predicted				

Confusion matrices of the best model (AlexNet) on the Raw, Filtered and Edge test sets.

5. Discussion

5.1 Which detector suffered the most from noise?

Canny was actually the most noise-sensitive of the five, even though it already has a smoothing step built in. Under Gaussian noise its F1 against the clean edge map fell to 0.33, and under salt-and-pepper noise it fell all the way to 0.04 (mean 0.19), with as many as 98% of its salt-and-pepper edge pixels turning out to be false. Ranked from worst to best (mean F1 vs. clean): Canny (0.19), Prewitt and Sobel (0.54 each), Laplacian (0.54), and LoG was the most stable (0.60). Textbook theory says that a detector with no smoothing and a higher-order derivative, such as the plain Laplacian, should be the noisiest, because noise lives in the high-frequency part of an image and a derivative amplifies high frequencies. Our results do not fully agree with that — the un-smoothed Laplacian was not the worst, Canny was. A likely reason is

that Canny's non-maximum suppression and double threshold are tuned for a fairly clean gradient; once noise pushes many pixels above the low threshold, the hysteresis step starts linking noisy pixels into long false contours instead of leaving isolated speckle behind, the way Laplacian does.

5.2 Did filtering help?

Gaussian filtering gave only a small average F1 gain for Gaussian noise (+0.02) and a bigger gain for salt-and-pepper noise (+0.09). Median filtering helped in both cases and by a much larger margin, especially for salt-and-pepper noise, where the average F1 gain across all five detectors was +0.43 (Laplacian alone improved by +0.46). This matches the usual expectation that a median filter is the natural match for impulse-type noise, since it simply drops the extreme “salt” and “pepper” pixel values instead of blending them into the neighbourhood the way a Gaussian filter does. For Gaussian noise, though, the median filter (+0.03) edged out the Gaussian filter (+0.02) slightly in our numbers, which is a bit different from the textbook rule that a Gaussian filter should be the better match for Gaussian noise — the gap is small enough that it could just be specific to this sample of 16 images.

5.3 What did the Canny threshold and kernel study show?

Moving from strict thresholds (100/200) to loose thresholds (30/100) multiplied the edge-pixel count roughly 8.3 times, from 143 to 1186 pixels per image, because many more weak gradients — skin texture, hair — get accepted. The Gaussian-noise F1 also shifted with the thresholds, from 0.28 at the strict setting to 0.16 at the loose setting, showing that a looser threshold is not just “more edges” but also “less reliable once noise is present”. Increasing the blur kernel from 3×3 to 5×5 at the 50/150 setting cut the edge count from 409 to 219 pixels and slightly lowered the noise F1 from 0.33 to 0.31. Considering both robustness and usefulness together, the winning configuration was Canny-5 (low = 30, high = 100, kernel = 5×5), and this was also the detector chosen for the whole classification stage.

5.4 Did the edge-only images help or hurt classification?

They hurt, clearly and consistently. All five models lost accuracy when trained on Set C instead of Set A, by an average of 22.2 percentage points (SVM and Random Forest: −25.0 pp each, KNN: −9.4 pp, ResNet50: −29.7 pp, AlexNet: −21.9 pp). This makes sense for a few reasons. An edge map throws away colour, which is one of the main cues used to tell melanoma apart from a benign nevus. It also throws away the smooth shading and inner texture of the lesion, while keeping unrelated “edges” such as hair and skin creases as if they were part of the lesion. As the figure above shows, several classes barely produced any detectable edge at all under the chosen Canny setting, so the models were sometimes working with almost no useful signal. Finally, both CNNs were pretrained on ordinary RGB photographs, so an edge map does not resemble anything their pretrained filters were originally built to recognise, and no amount of fine-tuning can put back information that was already deleted before training even started.

5.5 What information does an edge map lose?

An edge map only records where intensity changes sharply — it says nothing about colour (pigment colour and its variation, which matters a great deal for melanoma), overall brightness or shading (dark versus light lesions), the smooth texture inside the lesion, or which side of a boundary is “inside” versus “outside”

it. Once the map is turned into a binary black-and-white line drawing, even the strength of each edge is lost. What survives is essentially the shape of the border, which is only one part of the usual ABCD-style rules (asymmetry, border, colour, diameter) that guide how a lesion is assessed.

5.6 Why do CNNs learn better features than hand-made edge detectors?

The early layers of a CNN naturally learn small filters that look a lot like Sobel-style gradients and colour-contrast detectors on their own, so a hand-made edge map is largely redundant information for the network. A CNN goes further, though: it tunes the scale, direction and effective threshold of these filters to whatever the task actually needs, instead of a value chosen by hand (like the Canny thresholds in Task 3); it keeps using colour and texture at the same time as shape; it can combine edges into textures, then parts, then whole lesion patterns automatically through its deeper layers; and its filters are optimised jointly with the noise and augmentation the model will actually see, which makes them naturally more robust. Feeding a CNN a hand-made edge map instead of the original image mostly just removes information without giving the network anything it could not already learn to find on its own.

5.7 Which representation was best overall?

Averaged over all five models, the raw images gave the best results (58.4% accuracy, 55.9% macro-F1), the filtered images were almost identical (58.1% / 54.0%), and the edge images were far behind (36.2% / 35.1%). The single best run in the whole study was AlexNet trained on the raw images (67.2% accuracy, 65.7% macro-F1). It is worth keeping in mind that the test set has only 64 images, so a single image is worth about 1.56 percentage points of accuracy — a handful of misclassified test images can shift the ranking between models. For this reason, the validation numbers in Table 3b (Section 4.4) should be read alongside the test numbers rather than trusting the test score alone.

6. Conclusion

This lab showed that a detector's sensitivity to noise does not simply follow the rule “more smoothing means safer” — Canny, despite having a smoothing step, was the shakiest of the five detectors under both Gaussian and salt-and-pepper noise, while LoG (Gaussian blur followed by a Laplacian) was the steadiest. Median filtering clearly worked better than Gaussian filtering for salt-and-pepper noise, and the two filters performed closer to each other for Gaussian noise. The Canny threshold-and-kernel study confirmed the usual trade-off: loose thresholds catch more of the lesion border but also more clutter, while tight thresholds are cleaner but can miss weak boundary sections. The chosen setting, Canny-5 (30/100, 5×5), gave the best balance for this dataset among the six configurations tested.

For the classification stage, the main finding is that edge-only images are not a good input for this problem: accuracy fell by an average of 22.2 percentage points compared with the raw images, and every single model tested got worse, while the Lab-02 filtered images changed accuracy by only about 0.3 percentage points on average. The best-performing model overall, averaged across all three representations, was the AlexNet-based CNN (mean macro-F1 = 55.2%), just ahead of ResNet50 and SVM. Because the test set only holds 64 images, small differences between models should not be treated as the final word; repeating the

CNN runs with several random seeds and looking at the validation numbers alongside the test numbers would give a more reliable picture than a single test-set score.

7. References

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